

LAB 6 REPORT
A SHADOWGRAPH ANALYSIS OF BOOM XB-1, X-59, AND BOEING BIRD OF PREY

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1 INTRODUCTION

Reducing the sound associated with supersonic aircraft has been of keen interest to both civil and military aviation for many decades. Over the past decades, there has been a renewed interest in producing aircraft which can reduce the noise associated with the shockwave (Bonavolontà et al., 2023).

In this lab, three aircraft were analyzed which exhibit characteristics associated with low shock noise. For each aircraft, shadowgraph imagery was analyzed to understand the structure of shocks on each aircraft. The aircraft that are analyzed in the supersonic wind tunnel are the Boeing Bird of Prey, the NASA X-59, and the Boom XB-1.

The Boeing Bird of Prey is a demonstrator aircraft designed and built in the 1990s to test low observable stealth techniques. The project also tested new methods of aircraft design and construction such as using large single-piece composite structures among other things (National Museum of US Airforce, n.d.).

The NASA X-59 is designed to have less of a sonic boom and more of a sonic “thump” (NASA, 2025). This would mean that the aircraft is quieter, which is important as one of the biggest drawbacks of supersonic aircraft is the noise levels. The long nose on the aircraft makes this possible as it spreads out the supersonic shockwaves and prevents sonic booms (Doyle, 2021).

The Boom XB-1 is a test aircraft that was used to prove that the company, Boom, could design and build a supersonic aircraft. The aircraft paves the way for the company's commercial passenger aircraft, Overture (Robinson & Coles, 2025).

The X-59 and XB-1 are designed for supersonic flight, leading to the formation of shocks on the aircraft. It is important that engineers know where these shocks occur in flight, and putting a model in a wind tunnel at supersonic flows and looking at the shadowgraph images is a good way to see where and which types of shocks occur on the aircraft. Even though the Boeing Bird of Prey as tested did not achieve supersonic speeds, it has the design of a supersonic aircraft, meaning shadowgraph images will still convey useful information about where and what kind of shocks would occur on the aircraft.

Our testing is reliant on the use of a blowdown supersonic wind tunnel. Below is a diagram of a converging-diverging nozzle where Region I is subsonic and Region II is supersonic. The test section is in Region II.

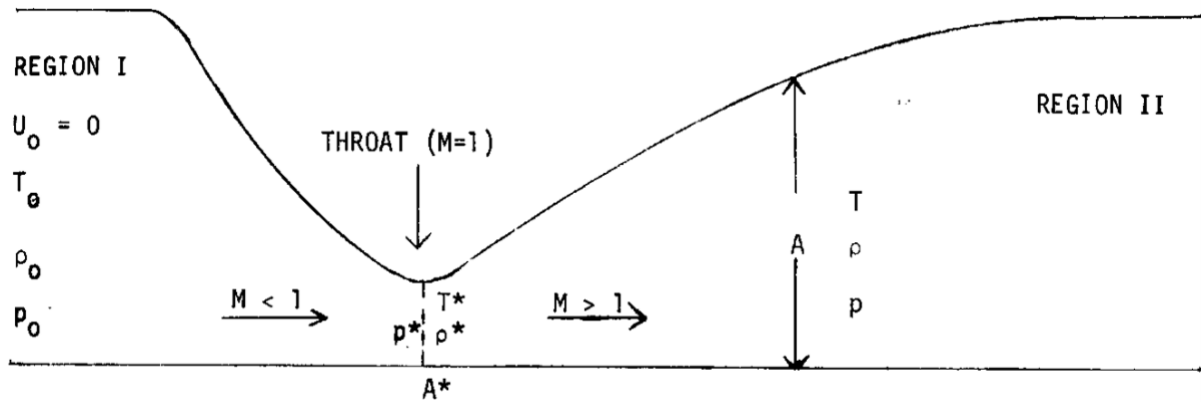


Figure 1: Converging-Diverging Nozzle Cross Section (Bane & Agrawal, 2025)

Absolute pressure (P_{abs}) is calculated by equation 1 below where P_a is atmospheric pressure and P_g is gage pressure.

$$P_{abs} = P_a + P_g \quad (\text{Eq 1})$$

Equation 2 below shows the calculation for the pressure differential ($\frac{P}{P_0}$) where P_{plenum} is the plenum pressure (absolute) and P_{abs} is absolute pressure.

$$\frac{P}{P_0} = \frac{P_{plenum}}{P_{abs}} \quad (\text{Eq 2})$$

Using the ratio of the static and stagnation pressure at each port, the Mach number at each section can be calculated to high accuracy.

2 OBJECTIVE

The experiment aimed to study the shock structures that form around three aircraft geometries representing both subsonic and supersonic regimes, specifically the Boeing Bird of Prey, the NASA X-59, and the Boom XB-1. By visualizing and comparing these configurations using shadowgraph imaging in the ARMS supersonic wind tunnel, we seek to understand how aerodynamic design influences the formation, strength, and distribution of shock waves. The primary objectives are: (1) to observe and qualitatively analyze the shock patterns produced by each aircraft model, and (2) to relate these observations to specific design features intended for stealth performance or supersonic noise reduction.

3 METHOD

3.1 Experiment Setup and Test Model(s)

In this lab, Supersonic Wind Tunnel A located in room B098 at the Neil Armstrong Hall of Aerospace Engineering at Purdue University was utilized. The supersonic tunnel is quite distinct from standard subsonic wind tunnels. The tunnel operates by blowing air from the building's shared high pressure supply line, which ranges from 90-130 psi; however, most tests occurred at 120psi. After opening the valve to the main supply, air flows into a settling chamber, known as a plenum or reservoir, so the known pressure can be treated as a stagnation pressure. Following the reservoir, air then flows through a converging-diverging section in which the Mach number is increased to attain a Mach number of 1 at the throat followed by a continued acceleration to supersonic flow in the test section. This tunnel has the capability to run at speeds near Mach 2.5, which is suitable for the purposes of measuring the aircraft in this experiment. The wind tunnel has a rectangular cross section with a constant width of 1" and a varying height along the tunnel's length. The tunnel is equipped with a shadow graph imaging system which uses a series of mirrors to focus a beam of light over the test specimen and into a lens to visualize compressible flow interactions. The tunnel uses a voltage control, specified in volts, to correspond to a plenum pressure to control the tunnel's speed.

Supersonic Wind Tunnel A is also equipped with a series of 17 pressure taps along the tunnel's length. Tap 1 is in the reservoir, Tap 3 at the throat, and Tap 15 at the start of the test section. Each pressure tap is connected to a pressure transducer to allow for data collection through the LabVIEW interface. Each pressure tap also has a known nozzle height which allows for an area of the tunnel at each pressure tap to be calculated. A diagram of the tunnel layout as well as a table of relevant pressure tap information are located in Figure 2 and Figure 3, respectively.

Operation of the Supersonic wind tunnel also requires following the relevant safety procedures. Before operating the tunnel, all room occupants must wear hearing protection and ensure the door is closed. During operation, no one may go behind the wind tunnel.

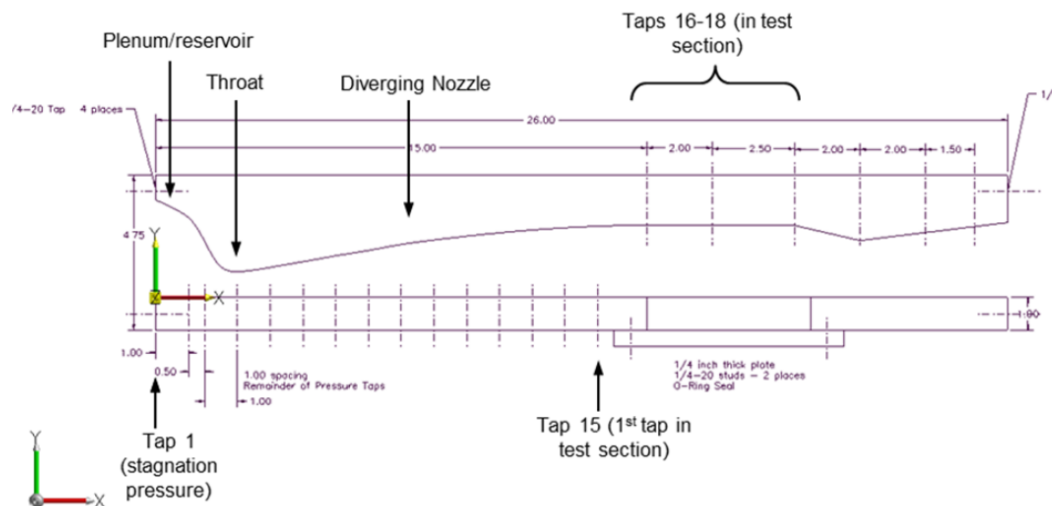


Figure 2: Supersonic wind tunnel diagram and pressure tap information

Tap Number	Distance from throat to pressure tap location (in)	Height of tunnel cross-section (in)	Pressure Scanner Channel No.
1	-1.5	2.455	16
2	-0.5	0.937	15
3	0	0.796	14
4	1	0.926	13
5	2	1.1	12
6	3	1.281	11
7	4	1.447	10
8	5	1.628	9
9	6	1.767	8
10	7	1.883	7
11	8	1.986	6
12 (not used)	9	2.074	Not used
13	10	2.14	5
14	11	2.189	4
15	12	2.213	3
16	13	2.213	2
17	14	2.213	1

Figure 3: Pressure tap relevant information

The test models for this lab consisted of three nose-sections, one for each of the analyzed aircraft. The full aircraft .STL files were originally sourced from Thingiverse for the XB-1 and Bird of Prey from users pershing1945 and HappyBattleSheep, respectively. The model for the X-59 was sourced from a public NASA project for 3D printing the plane. The original 3D models are depicted below in Figure 4, Figure 5, and Figure 6.



Figure 4 : Boom XB-1 3D Model



Figure 5 : X-59 3D Model



Figure 6 : Boeing Bird of Prey 3D Model

In an ideal scenario, each model would be scaled to fit in the entirety of the supersonic test section area, which has a fuselage length limit of 1.8" and a wingspan limit of 1". After scaling the models to fit into the tunnel, two limiting factors became apparent. First and foremost, some of the smaller geometric features, such as the spike on the front of the XB-1 and the thin wings, had total thicknesses of ~0.2-0.5 mm that resulted in only a few layers of filament when printed using a layer height of 0.12 mm. The second limiting factor was the mounting of the test models into the tunnel. Models are mounted by being threaded onto a stud which was located at the aft of the model. A picture of the mounting block is pictured below in Figure 7.

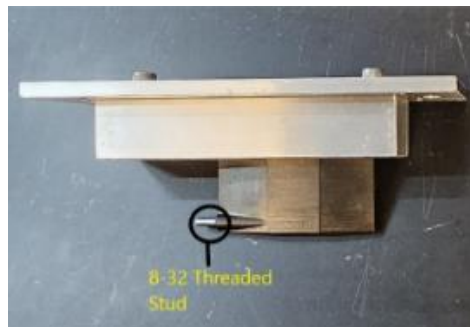


Figure 7 : Mounting Block for the Models (Bane & Agrawal, 2025)

For models to be mounted into the supersonic tunnel, an untapped hole of diameter 0.137" with a minimum depth of 0.5" must be located at the back of each model. For the full-aircraft models which had rear geometry less than the required hole diameter, the addition of a cylinder with length 0.591" and diameter 0.297" was added to sit flush with the most rear point of each aircraft. In order to reduce the effects of the flat cylindrical face, a cone of length 0.220" was added in front of the cylinder. This served as structural support for mounting hardware, which allowed full-aircraft models to be placed into a test section of the tunnel. CAD models for the modified aircraft are below.

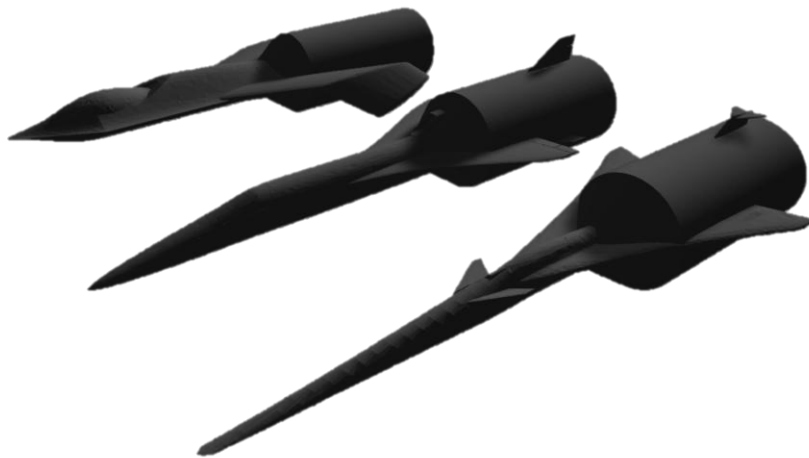


Figure 8 : Full Body Section of the Bird of Prey (left), XB-1 (middle) and X-59 (right)

In addition to the full-aircraft models, each aircraft's respective nose cones were also fabricated as test models. Unlike the full-aircraft models which were scaled down to the size constraints of the tunnel, for the nose models, each .STL file was scaled until the required mounting geometry fit inside the model without requiring any additions to the model itself. CAD models for the nose sections are located below.

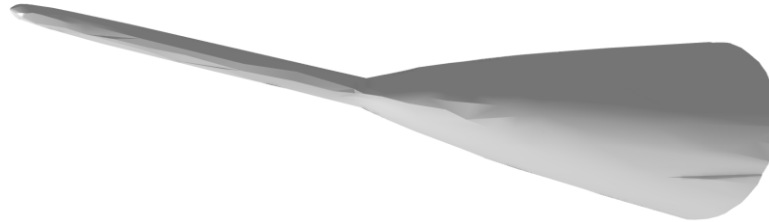


Figure 9: Nose Section of Boom XB-1

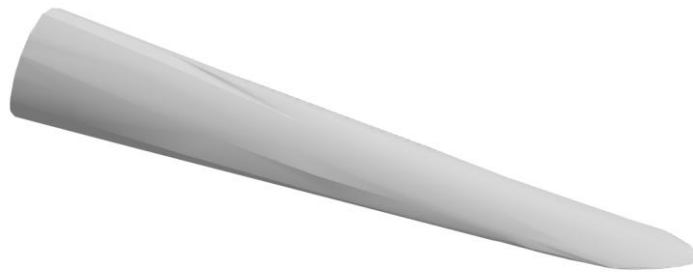


Figure 10: Nose Section of X-59



Figure 11: Nose Section of Boeing Bird of Prey

Each of the models was manufactured via 3D printing. Specifically, all models were printed with PLA using a layer height of 0.12mm. Without losing feature fidelity in the slicing, all the models were printed “belly-side-down” on the print bed. This required incorporation of tree-style supports. All models were printed using 100% infill to maximize survivability in the wind tunnel. To ensure as smooth of a surface as possible to limit unintended shocks, each model (specifically the side with supports) was sanded via three different grit sandpapers. Once the models were printed and post-processed, the mounting hole for each model was tapped and test-fit with the mount. The resulting models pictured below are the ones which ended up as the final test models. Note some were printed in white PLA and some were printed in black PLA due to running out of black PLA during manufacturing.

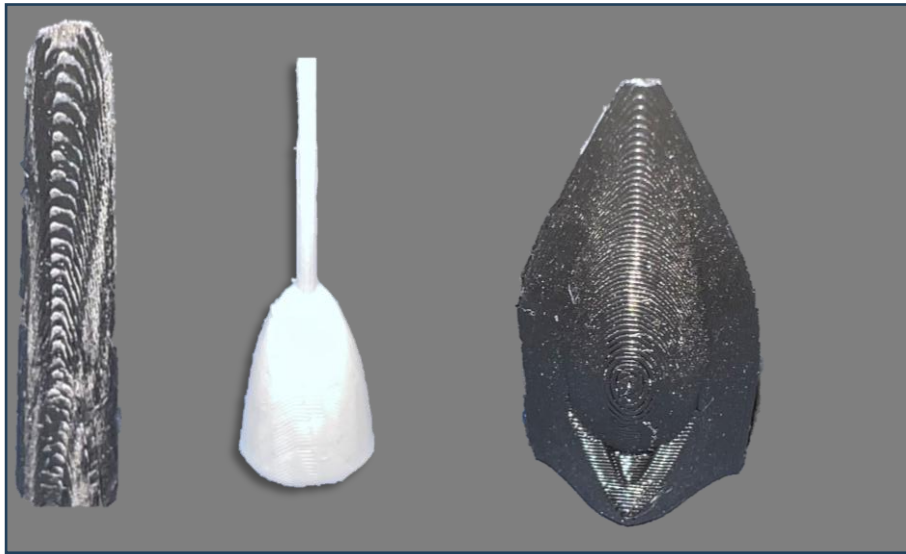


Figure 12: Nose cone test models for X-59 (left), XB-1 (middle), and Bird of Prey (right)

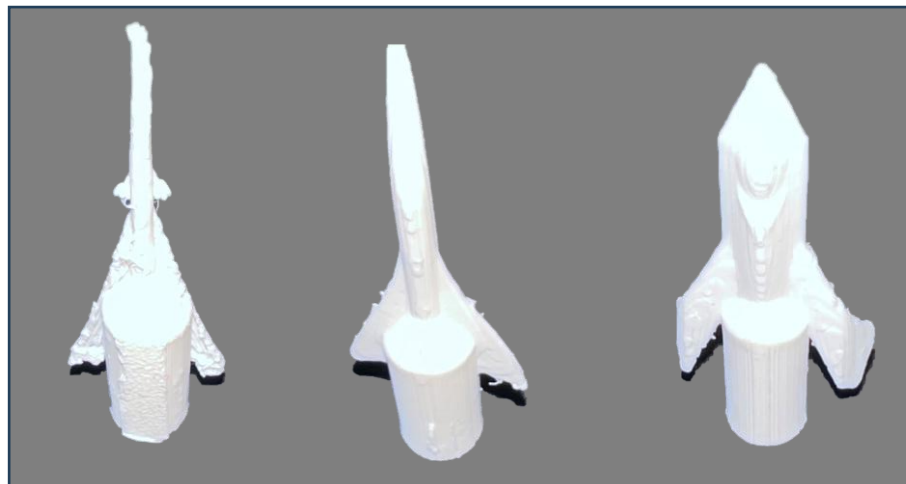


Figure 13: Full aircraft test models for X-59 (left), XB-1 (middle), and Bird of Prey (right)

3.2 Instrumentation

The main instrument used for this lab was the shadowgraph imager on the supersonic wind tunnel. Each image was recorded with a $5\ \mu\text{s}$ exposure, and a video sequence recorded at 1000×760 , 30 fps was captured. This video sequence was used to get images of the shocks for each of the models. This camera had no significant lack of temporal or spatial resolution, so it was adequate for measurements and represented little to no loss in accuracy of measurement.

In addition to the shadowgraph imager, the tunnel included a digital pressure transducer system which measured the static pressure at fixed locations along the tunnel. The pressure transducer system used the PSI 9000 Series pressure transducers with an ADC for readout in LabView. In experiments, these PT measurements fluctuated on the order of 1-5% of the stagnation pressure. However, the pressure measurements were purely supplementary for our purposes, and this measurement error had little impact on the attained results.

3.3 Experimental Conditions

The tests were conducted using an approximately constant plenum pressure of 50 psia.

Table 1: Supersonic Tunnel Operating Conditions ~50 psia Plenum Pressure

	Plenum Pressure psia	Atmospheric Pressure psia
Control	50.686	14.66
Boom XB-1	50.152	14.66
Bird of Prey	50.152	14.66
NASA X-59	50.986	14.66

The Mach number in the test section can be calculated using the ratio of static to stagnation pressure since there are no shocks leading up to the test section (otherwise the test section would not yield shocks). This relationship at Tap 15 (start of test section) was used as the input to an isentropic flow compressible calculator (Devenport, 2023) to find the Mach number for each scenario.

	P/P0	Mach #
Control	0.075143505	2.3398
Boom XB-1	0.075010865	2.3409
Bird of Prey	0.075480655	2.3370
NASA X-59	0.075342736	2.3381

3.4 *Procedures*

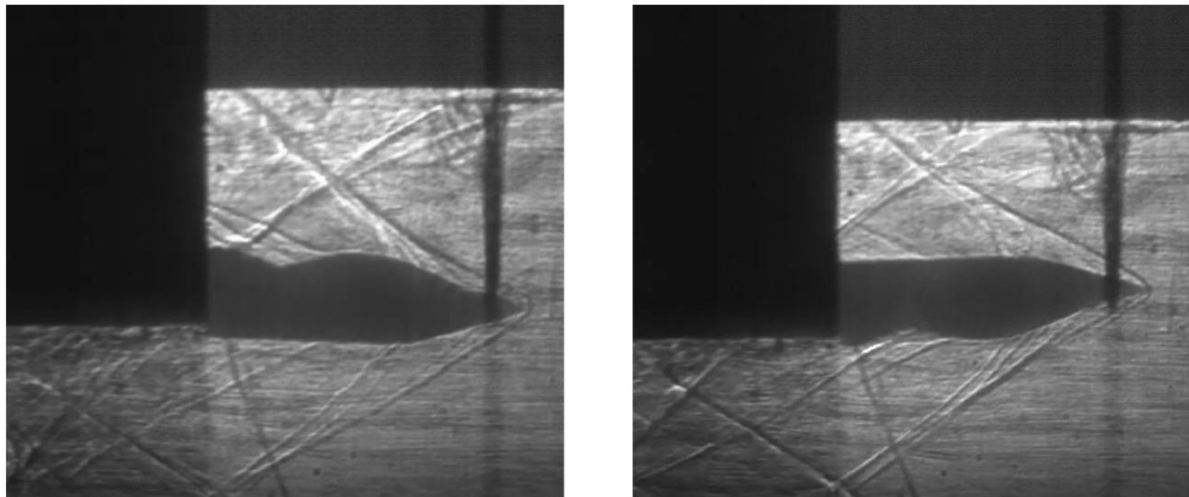
The testing setup was similar to that used in Lab 4. To start, the pressure sensor was turned on and the LabVIEW software was opened. The valve upstream of the wind tunnel was opened, and the supply pressure did not drop below 90 psi during the test (if it did, the test would have been aborted). The Nozzle Reservoir/Plenum Pressure was assumed to be the stagnation pressure and was recorded in psia. A file was created and synched to LabVIEW to store the data that was collected from the tunnel. The LabVIEW program was run with the tunnel not turned on to see if the data saved correctly when “push to write” was selected. This button was pushed for each subsequent test to record the gage pressure data.

To test the control, the tunnel was clear of a specimen. Before the tunnel was turned on, everyone in the room was wearing ear protection. The tunnel was then turned on to a plenum pressure of 50 psi and the pressure data from LabVIEW was recorded from 16 taps along the tunnel. The data was then copied from the data file to an Excel Spreadsheet where it was organized.

After the control test was completed, the first model was installed in the test section. Before the tunnel was turned on, everyone in the room was wearing ear protection. The tunnel was turned on to the same plenum pressure as the control test (50 psi), and shadowgraph images and pressure measurements were recorded. The images came from a screen recording that was taken from the camera that was looking at the test section. The data was also copied from the data file to the same Excel Spreadsheet where it was organized. This procedure was repeated for each model that was present.

4 RESULTS

Below, the shadowgraph images are presented for each of the aircraft in our analysis. The first aircraft, the Bird of Prey, was tested in two orientations. The Figure below depicts the two orientations of the model as tested in the ARMS supersonic wind tunnel.



(a) Bird of Prey (Standard Orientation)

(b) Bird of Prey (Reversed Orientation)

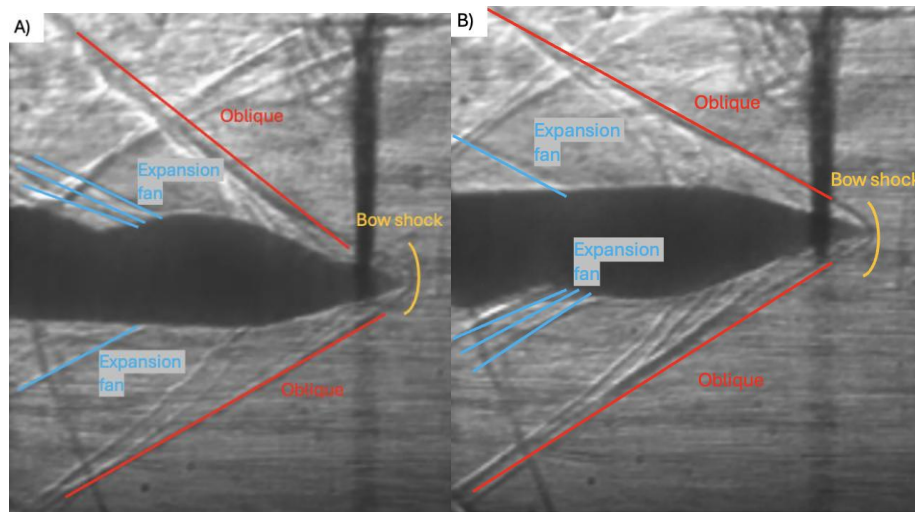


Figure 10: a) Bird of Prey in standard orientation, showing strong oblique shock on aircraft nose. b) Reversed orientation of Bird of Prey, showing more complex shock structure on the lower side, especially around the inlet. Each with a respective highlighted plot of shock formation

The model showcases many interesting shock structures. Notably, the inlet of the aircraft (located right behind the ‘hump’ of the cockpit) has an interesting shock structure which seems to create a smoother flow into the inlet. The shock structures surrounding the nose of the aircraft are more reminiscent of a typical cone or wedge shape.

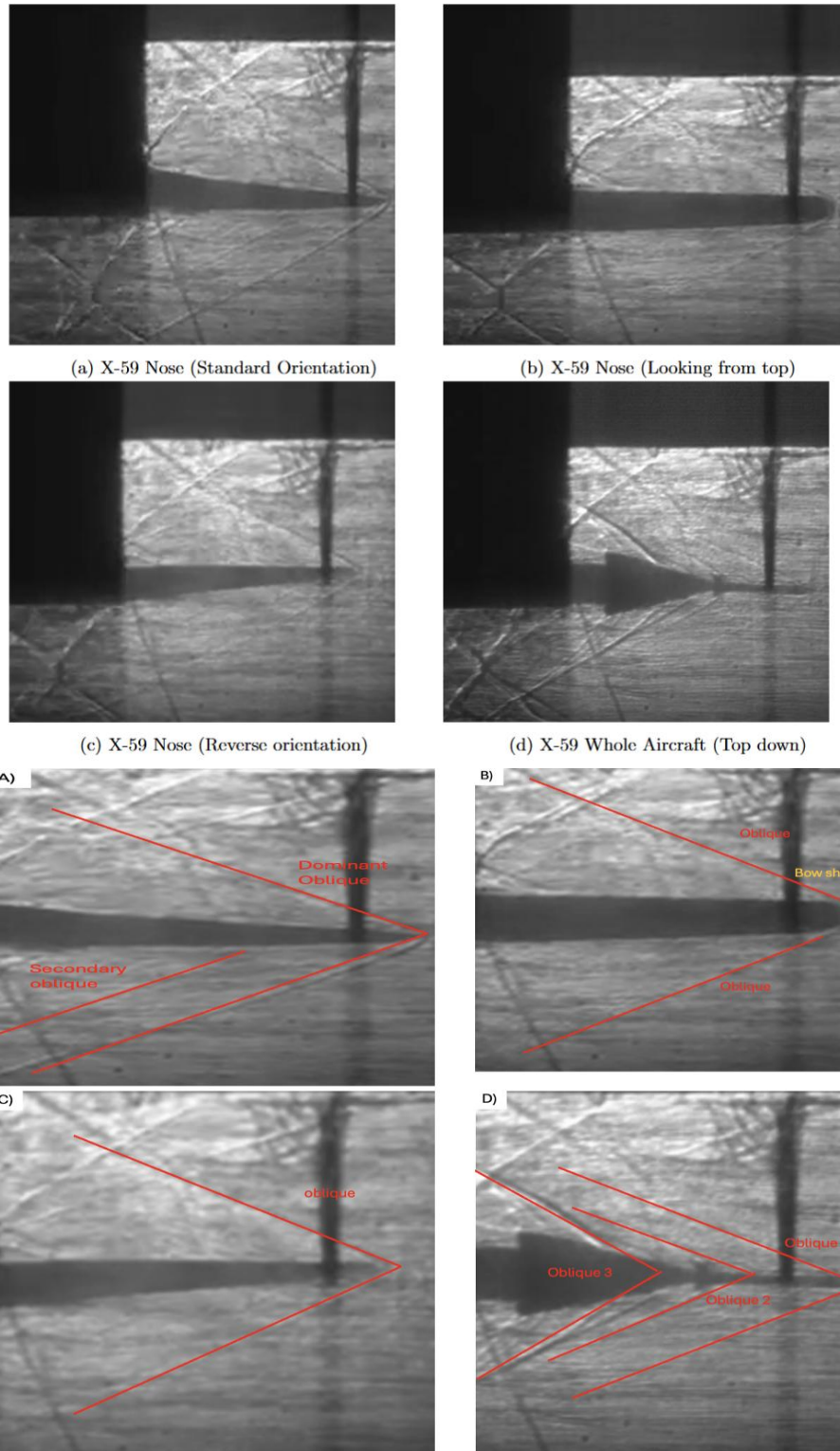


Figure 11: a) Standard orientation of X-59, showing an oblique shock at the front. b) X-59 Nose from a top down view, showing very weak shocks from this angle. c) X-59 nose in the reverse orientation, showing a similar shock structure to the standard orientation. d) A view of the whole X-59 from a top down view, showing the complex shock structure coming from the canards and wings of the aircraft. Each view in the figure has its own expansion/shock labeling.

Looking at the shock structure of the X-59, the elongated nose seems to produce multiple smaller conical shocks, starting at the very tip, again at the smaller canards, and a series of shocks along the thicker region of the fuselage where the wing box is located. This broken-up series of shocks likely contributes to reducing the volume of the sonic boom and to cleaning the flow before the inlet located at the far back of the plane. As compared to the Bird of Prey, the shock structures of the X-59 appear much more faint on the shadowgraph. Looking from the top, the shock structures are very weak at the front of the nose.

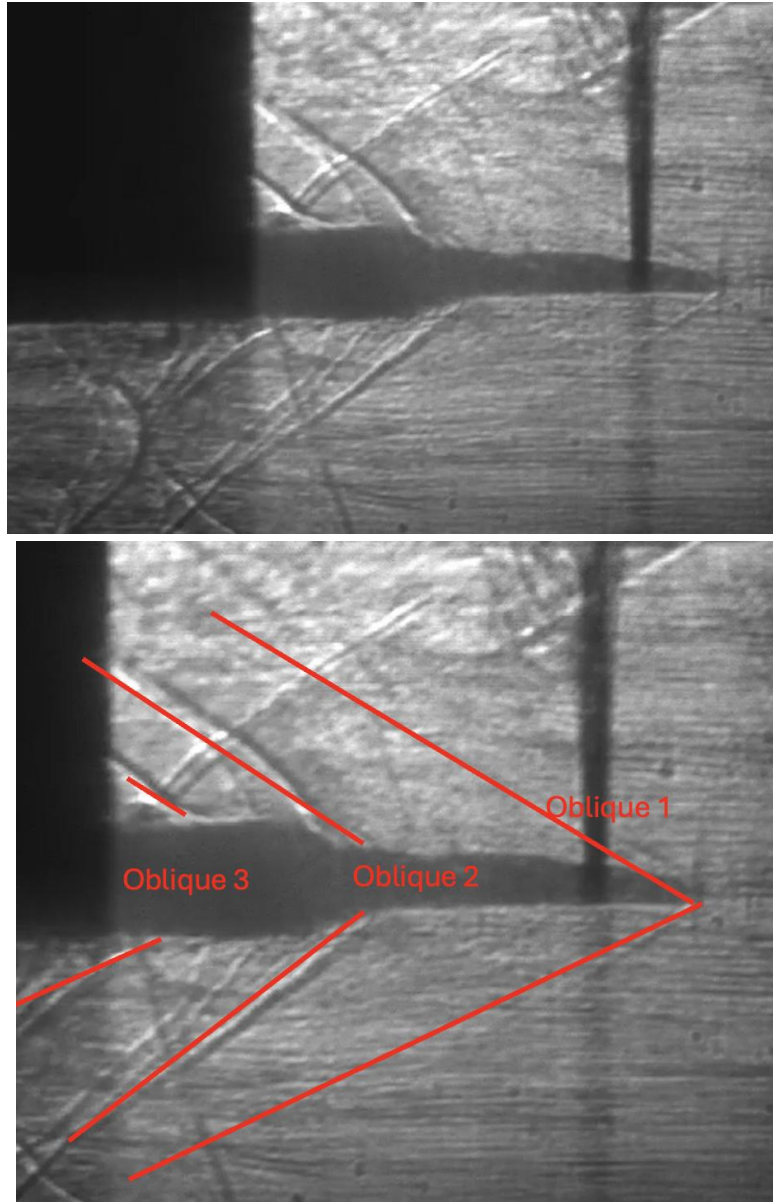


Figure 13: Flow around the nose of the XB-1. The nose of this aircraft is nearly axisymmetric, and shocks look similar in all directions. The front oblique shock (oblique 1) is quite weak, and stronger shocks are seen with the area increase at the rear of the aircraft (oblique 2 and 3).

5 DISCUSSION

The objectives of this experiment were to study the shock patterns produced by each model and relate these observations to design features intended for stealth performance or supersonic noise reduction. The shock patterns from the design of each aircraft are analyzed in the previous section.

One limitation of the experiment is that the printed models did not replicate the aircraft precisely. Due to the small scale required for testing in the supersonic wind tunnel, the resolution of the 3D printer was not high enough for exact models. The general structure of the aircraft were clear, so shock patterns were visible and mostly easy to distinguish around the leading edge and most details. However, in the regions closest to the surface of the model, shocks were difficult to distinguish. This is likely due to the surface roughness of the models.

Several of the models' features were very thin, causing them to flutter during tests. The models themselves wobbled when the flow velocity was changed at the beginning and end of each test. During this motion, the shocks were still visible but became fainter with the oscillations of the model. This did not strongly affect the visualization of shock patterns. Since videos of each trial were captured, it was possible to obtain images where the shock structure was not compromised by the movement of the model. The models also remained intact, so images were taken during multiple tests at different angles.

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